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# Madden-Julian Oscillation Impacts on Tropical African Precipitation

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## Abstract

The Madden-Julian Oscillation (MJO) is the dominant mode of sub-seasonal climate variability in the global tropics. As such it represents an opportunity for intra-seasonal rainfall prediction and, perhaps, for explaining dynamics that underlie longer term variability and trends. This opportunity is of substantial interest for tropical Africa, where rainfall variability has significant impacts on agriculture, energy, and natural ecosystems. The objective of this review paper is to inventory and assess the state of knowledge of MJO influence on African rainfall. A number of studies have identified statistical links between MJO and sub-seasonal rainfall variability in West, East, and Southern Africa. The proposed mechanisms to explain this influence differ by region and by season, and they often involve multiscale interactions between local precipitation processes and MJO-associated atmospheric dynamics. Dynamically-based forecast systems have some skill in predicting MJO evolution to time horizons of 3-4 weeks, and some can capture teleconnections to Africa. On longer time scales, there is evidence that MJO activity both modulates and is modulated by the El Niño Southern Oscillation and the Indian Ocean Dipole. The implications of these interactions for MJO connections to Africa require further research, as does the potential for trends in MJO behavior and impacts on Africa under global climate change.


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## 1. Introduction

Sub-seasonal climate variability, including dry spells and periods of unusually high rainfall, can have significant impacts on ecosystems, agriculture, hydrology, and human health. The sub-seasonal timescale is also particularly relevant for decision making, as actions to support water resource allocations, disaster risk reduction, and crop management often take place on a time horizon of weeks to months. This climate services opportunity has motivated significant efforts on sub-seasonal to seasonal (S2S) forecasts with dynamical models (Vitart et al. 2012; Brunet et al. 2010; Kirtman et al. 2014), focusing on problems of forecast initialization, land-ocean-atmosphere coupling, and, notably, improved representation of the Madden-Julian Oscillation (MJO). The MJO has long been recognized as the largest mode of sub-seasonal climate variability in the tropics, and its impacts extend globally (Madden and Julian 1994). The MJO also has inherent predictability, in that the core convective region propagates eastward over the course of several weeks at a known range of speeds (Zhang 2005). Nevertheless, predicting the initiation and propagation extent of the MJO is a significant challenge, as is quantifying, understanding, and simulating its teleconnections beyond the convective core (Lau and Waliser 2012).

Tropical Africa, defined here in the traditional sense of the tropics ($23.5^0$S to $23.5^0$N), and understood as the region in which seasonality of precipitation is determined by the migration of the Inter-tropical Convergence Zone (ITCZ), includes tremendous climatic diversity. The Congo Forest and Gulf of Guinea coast are some of the wettest places on Earth, with average annual rainfall exceeding 2,000mm, while parts of the Sahara Desert to the north and the Kalahari and Namib Deserts to the south are hyper-arid, with annual rainfall in the range of 100mm or less. More importantly for the purposes of this review, tropical Africa exhibits diverse patterns of climate variability. This is true seasonally, as northern Tropical Africa receives rain in boreal summer, southern Tropical Africa receives rain in boreal winter, and equatorial Africa receives

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two rainy seasons, as the ITCZ passes from south to north and north to south. It is also true inter-annually, as regions respond differently to slowly varying climate modes. For example, El Niño is associated with boreal summertime drought in the Sahel and interior East Africa, while La Niña leads to drought in the spring (March – May) and fall (October – December) rainy seasons in the Horn of Africa (Nicholson and Kim 1997). Recognizing this diversity, the review is organized according to three broad climate regions associated with distinct seasons and synoptic dynamics: West Africa (spring and summer rains), Equatorial East Africa (fall and spring rains), and Southern Africa (boreal winter rains) (Figure 1). In the case of Southern Africa the review includes studies that extend south of 23.5$^{0}$S, as the literature for this region includes consideration of tropical to temperate dynamics.

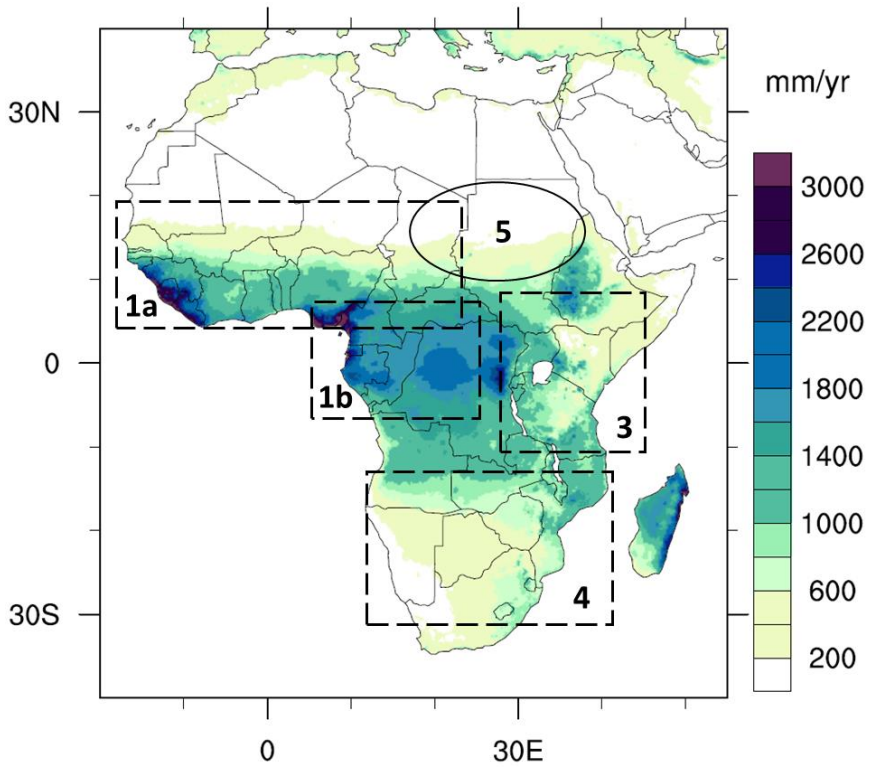


*Figure 1: Annual average rainfall in Africa, 1981-2014, based on Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS; Funk et al., 2014). Boxes approximately indicate the regions reviewed: (1a) West African Monsoon region; (1b) Equatorial West Africa; (3) Equatorial East Africa; (4) Southern Africa. Boxes are approximate because studies differ in their exact definitions. The oval (5) shows the triggering region for African Easterly Waves.*

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In all three regions MJO is shown to have significant impacts on intra-seasonal rainfall variability. But the mechanisms of influence are diverse: in some cases MJO-associated wave activity appears to be the dominant connection, in others the direct influence of the MJO convective core is more important, while in others MJO is thought to trigger synoptic dynamics that have a downstream impact on rainfall. Characterizing and, ultimately, predicting MJO impacts could be particularly valuable for regions of tropical Africa, which are home to some of the most climate vulnerable populations in the world (McMichael et al. 2008; Brown et al. 2007). Improved MJO prediction could be applied to food security interventions, disaster risk reduction, and, with sufficient operational decision support, improved crop and resource management (Coughlan de Perez et al. 2015; Thiaw and Kumar 2015).

This review is motivated by the growing body of literature that addresses the magnitude and mechanisms of MJO impacts on tropical Africa. It is intended to complement recent reviews of MJO dynamics (Wang 2012; DeMott et al. 2015; Li 2014) and to expand on an overview of MJO impacts on Africa and West Asia provided by Barlow (2012). This literature on MJO and Africa has, quite understandably, focused on mechanisms of sub-seasonal variability. At the same time, MJO interacts with modes of longer term climate variability, both as a potential trigger for large scale climate modes like the El Niño Southern Oscillation (ENSO) (Zhang 2005) and as a mediator of climate teleconnections (Hoell et al. 2014). Important trends in rainfall, including the recent decline in the East African long rains, can also be studied in terms of accumulated changes in the frequency of sub-seasonal wet and dry periods (Vigaud et al. 2016). In these ways, MJO sub-seasonal variability can aggregate to influence climate on longer time scales (Figure 2), with potential influences on inter-annual and long-term climate variability in tropical Africa. For this reason the review includes available literature on the manner in which MJO interactions with ENSO and other climate modes project onto tropical Africa. (Barlow 2012)

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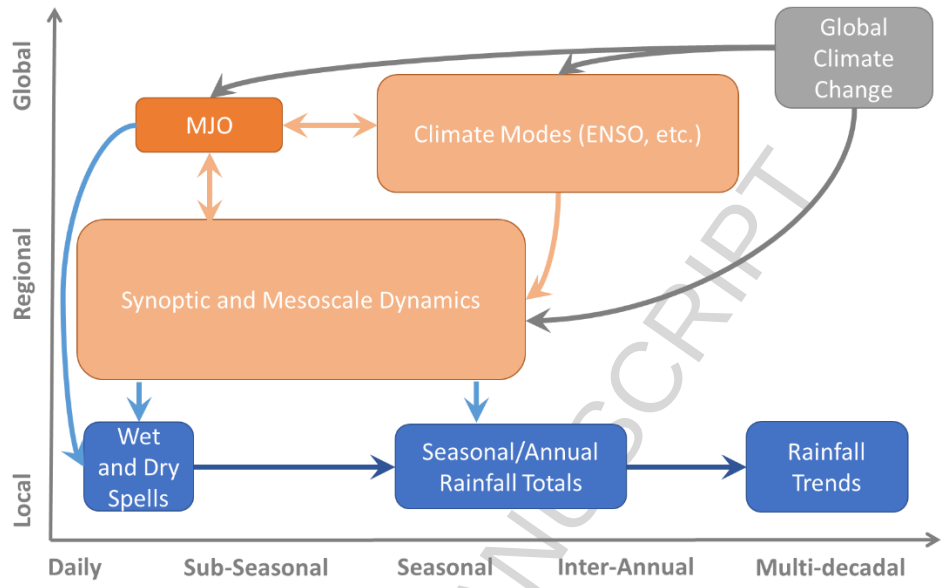


*Figure 2: MJO interactions across spatial and temporal scales. The pan-tropical MJO phenomenon influences short-term local rainfall variability through direct convective activity or, more commonly, through modification of synoptic and mesoscale dynamics. It also interacts with hemispheric to global scale climate modes, with phenomena like ENSO influencing and being influenced by MJO activity. Through these interactions MJO can influence rainfall totals on seasonal and longer time-scales. Global climate change acts primarily as a long-term external driver of change in dynamics across spatial scales.*

The remainder of the paper is organized as follows. A basic overview of MJO patterns and dynamics is provided in Section 2. In Section 3 MJO impacts are reviewed in detail for West Africa, Equatorial East Africa, and Southern Africa. Section 4 cross-cuts the regional dynamics described in Section 3 in order to summarize mechanisms through which MJO has been found to influence African rainfall. This leads to a brief review of associations between MJO and large scale modes of climate variability in Section 5, with an emphasis on ENSO. As the relationship between MJO and other modes of Pacific and Indian Ocean climate variability are widely studied, the review presented here is intended to provide a link to that literature rather than a comprehensive assessment. Section 6 covers applications of MJO to S2S forecasting efforts for Africa. Finally, Section 7 offers perspectives on future research directions.

## 2. The Madden-Julian Oscillation

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The MJO is a tropical disturbance with a characteristic timescale in the range of 30-60 days. It is responsible for the majority of weather variability in the global tropics (Madden and Julian 1994). MJO events typically present as a large-scale convective system that arises in the Indian Ocean, propagates eastward to the Western Pacific Ocean at a speed of ~5 m $s^{-1}$, loses convective coupling and propagates at higher speed through the cooler Eastern Pacific Ocean, and sometimes reemerges as a convective feature over the tropical Atlantic Ocean and Equatorial Africa (Madden and Julian 1971; Zhang 2005; Madden and Julian 1994). The MJO is strongest in boreal winter and spring and has a secondary peak in boreal summer (Hendon and Salby 1994; Zhang and Dong 2004), but it affects precipitation in both tropical and extratropical regions year-round (Donald et al. 2006; Zhang 2013). As noted by Zhang (2005), despite its name MJO is not actually a regular "oscillation," but rather an episodic mode of intra-seasonal variability. Time between active phases, convective strength, and speed of propagation vary from event to event.

The characteristics and dynamics of the MJO have been thoroughly reviewed by Zhang (2005), with updated review of specific topics by Wang (2012), Li et al. (2014) and DeMott et al. (2015). Briefly, the "active phase" of the MJO is characterized by a convective core composed of a large number of small scale deep convection systems. Zonal winds are convergent to the core near the surface and divergent aloft (Figure 3). Active phase convergence is dynamically linked to a "suppressed phase" of weak convection and surface divergence that propagates along with the active phase, producing a convection dipole that can span large portions of the global tropics. Large scale tropospheric heating associated with the MJO active phase gives rise to eastward-propagating dry Equatorial Kelvin waves and westward-propagating Equatorial Rossby Waves. Negative heating anomalies associated with the suppressed phase also give rise to this wave pattern, with opposite sign. These equatorial waves can play a critical role in linking MJO activity to African precipitation, particularly in West Africa, which is remote from the MJO's favored convection locations in the Indian Ocean and Western Pacific Ocean.

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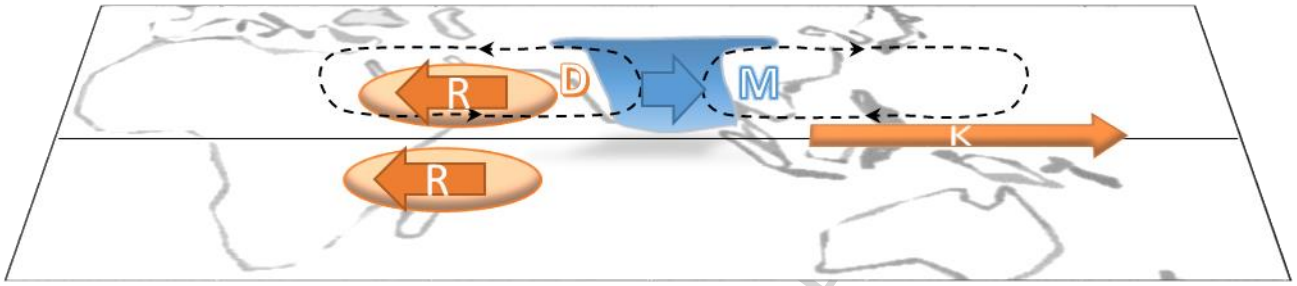


*Figure 3: Schematic of the MJO. The convective core (blue) propagates eastward at ~5 m s$^{-1}$. Equatorial Kelvin Waves (K) and Equatorial Rossby Waves (R) are both generated by MJO condensational heating anomalies. Kelvin waves propagate east at phase speed on the order of 30 m s$^{-1}$. Rossby Waves propagate west at speeds on order of 10 m s$^{-1}$. Surface winds converge at the convective core, while elevated winds diverge. The air in front of the MJO convective core tends to be anomalously moist (M), while air behind the convective core is anomalously dry (D).*

The formation and slow propagation of the MJO convective core has been the subject of extensive research, but no single theory has yet explained all of the observed features of the phenomenon. An early hypothesis was that MJO events are the product of stationary intra-seasonal oscillations (Yasunari 1979), but statistical analysis has not supported this mechanism, and simulated stationary oscillations fail to generate a propagating MJO (Yamagata and Hayashi 1984). Other relatively early studies of MJO dynamics focused on Wind-Induced Surface Heat Exchange (Matsuno 1966), in which enhanced wind speed to the east of a tropical convective core could enhance evaporation and favor eastward movement of the convective center (Emanuel 1987; Neelin et al. 1987). But observations of surface winds during strong MJO seasons are inconsistent with this theory. Moisture convergence associated with wave-induced Conditional Instability of the Second Kind (wave-CISK) has also been proposed as an MJO propagation mechanism (Lau and Peng 1987; Chang and Lim 1988), but this theory does not explain observed MJO scale and wave speeds. Adding friction to the wave-CISK model (Wang and Rui 1990; Wang 1988) improves prediction of these characteristics in models that allow for nonlinear heating (Li 2014).

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Other studies have focused on the MJO's modulation of synoptic dynamics as a mechanism for propagation (Biello and Majda 2005; Zhou and Li 2010), or focus on the role of moisture sources to the convective column (Sobel and Maloney 2013; Raymond and Fuchs 2009) and large scale moisture asymmetry east and west of the convective core (Hsu and Li 2012). Tropical stochastic forcing (Salby and Garcia 1987), midlatitude drivers (Lau and Peng 1987), radiation feedbacks (Hu and Randall 1994), and air-sea coupling (Wang and Xie 1998; DeMott et al. 2015; Li et al. 2008) have also been investigated as primary or supporting explanations of MJO propagation dynamics. These multiple theories for the MJO are not mutually exclusive. Some combination of processes could well be at work, and it is also possible that the relative importance of mechanisms varies by season and by whether an MJO event is primary (i.e., there was no immediately preceding MJO event) or is a secondary response to a previous MJO event (DeMott et al. 2015; Li 2014; Matthews 2008).

From a diagnostic perspective, MJO activity is typically described in terms of the strength and location of the convective center of action. The Real-time Multivariate MJO (RMM) index system (Wheeler and Hendon 2004) describes these features by taking the first two EOFs of combined daily mean fields of outgoing longwave radiation (OLR) and 850- and 200-hPa zonal winds averaged over $15^0$N–$15^0$S. These EOFs form the basis of a phase space diagram in which eastward propagating MJO events move counterclockwise while variability in the absence of MJO appears as random movement near the origin (e.g., Figure 4). RMM is the most widely used MJO indexing system, but similar indices have been derived on the basis of OLR alone (Kiladis et al. 2014) and circulation anomalies defined by zonal winds and velocity potential (Ventrice et al. 2013). Some researchers also choose to define a customized MJO index that is optimized to capture MJO-related convection variability in their study region (Waliser et al. 2008; Berhane et al. 2015; Pohl et al. 2007). These indexing systems are valuable for MJO studies, as they conveniently define the key characteristics of MJO activity for use in diagnostic studies of dynamics and impacts. At the same time, it is useful to remember that the indices are

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imperfect simplifications of MJO activity. They bin a continuum of MJO expressions into discrete categories that may not capture the full diversity of MJO behavior. Analyses of MJO activity can be sensitive to choice of index (Kiladis et al. 2014), and it is possible for teleconnection signals to be lost or aliased by the pre-defined bins. Alternatives to standard indexing systems have been proposed in order to capture event-to-event variability in the MJO (Goulet and Duvel 2000; Bellenger and Duvel 2012; Pohl and Camberlin 2014). Broader use of these approaches could enrich understanding of MJO dynamics, and in particular the interaction of MJO with other modes of climate variability.

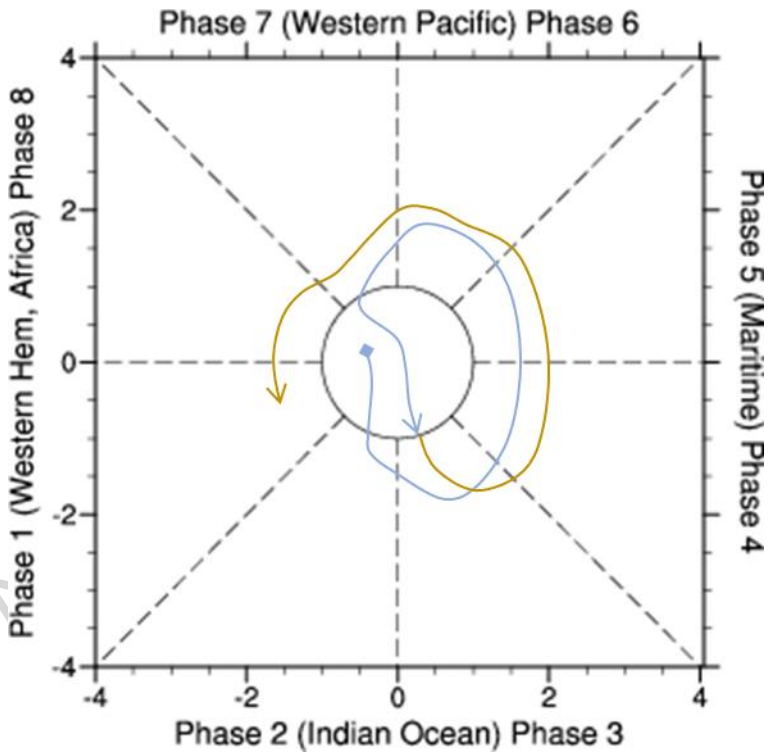


*Figure 4: RMM phase space diagram, showing an idealized example of two MJO cycles. The first (blue) arises in the Western Indian Ocean and propagates to the Western Pacific before becoming indistinct. The second (gold) arises in the Eastern Indian Ocean and propagates through to the Western Hemisphere. Such a pattern would typically evolve over 2-4 months.*

## 3. MJO Impacts on Africa

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As described in Section 2, the MJO can influence dynamics relevant to precipitation in tropical regions through direct enhancement or suppression within the MJO convective core, by modifying synoptic atmospheric dynamics, and via the influence of Rossby and Kelvin waves that emanate from the MJO convective core. Each of these must be considered when evaluating the potential for MJO to influence precipitation in regions of tropical Africa. Seasonality is also a key consideration. Since MJO is strongest in boreal winter, there is reason to expect a strong precipitation influence in southern Africa, which experiences its precipitation maximum during those months. However, MJO activity is still pronounced during transition seasons of spring and fall, when rainfall is maximal in equatorial West and East Africa. In boreal summer the summertime MJO (sometimes referred to as the "Intra-Seasonal Oscillation" (ISO)) influences monsoon dynamics in Asia, and it has the potential to do the same during the critical summertime rainy season in northern tropical Africa, including through influence on the West African Monsoon. The following sections review the state of scientific literature for each of these regions and seasons.

## 3.1 West Africa

West Africa, defined here to include both Equatorial West Africa and the West African Monsoon zone, has been the subject of a number of MJO-oriented studies of sub-seasonal climate variability. The majority of these have focused on the July to September summertime rainy season, when the West African Monsoon is active and when the Sahel receives the majority of its annual precipitation. Within this set of studies, there has also been specific examination of the influence that MJO has on African easterly waves, which are relevant to precipitation within the region and to cyclogenesis downstream. Fewer studies have examined springtime variability, which is critical for Equatorial West Africa, including the Gulf of Guinea coast.

### *3.1.1 Summertime West African Rainfall*

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Studies of MJO influence on summertime rainfall in West Africa emerge from the observation that rainfall during the West African Monsoon (WAM) exhibits two modes of systematic intra-seasonal variability. The first, higher frequency mode of variability has variously been described as 10-25 day, 6-30, or 15 day variability (Sultan et al. 2003; Janicot and Sultan 2001; Kiladis and Weickmann 1997). Proposed mechanisms of variability on this time scale include modulation of zonal moisture flux into West Africa and meridional shifts in the ITCZ (Mounier and Janicot 2004).

The second, lower frequency mode of variability peaks in the 25-60 day time range (Janicot and Sultan 2001). Early global studies of MJO convective signals did not identify the connection between this WAM variability and the MJO (Knutson et al. 1986; Murakami et al. 1986), or found only a weak signal (Knutson and Weickmann 1987). Matthews (2004), however, isolated the relevant mode by applying a 20-100 day bandpass filter on outgoing longwave radiation (OLR) fields for West Africa and surroundings (10S-20N; 40W-40E) and identifying the leading eigenvector from empirical orthogonal function (EOF) analysis. This leading eigenvector was found to account for 15.8% of variance over the period of analysis, far exceeding the second and third eigenvectors. Negative OLR anomalies associated with this mode (i.e., enhanced convection) were found to lag anomalously low convection in the Western Pacific equatorial warm pool by 20 days, providing an observational basis for a link between WAM variability and the MJO center of action.

Matthews (2004) proposed a mechanistic link between the two regions based on dry Kelvin and Rossby wave propagation. Negative convective anomalies in the Western Pacific, associated with the MJO suppressed phase, result in reduced condensational heating and a net negative atmospheric heating anomaly over the equatorial warm pool. This negative heating anomaly generates an eastward propagating dry Kelvin wave and westward propagating Rossby wave that converge over West Africa 20 days later, bringing negative mid-tropospheric temperature

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anomalies that destabilize the atmosphere and favor deep convection in the WAM region. The Kelvin and Rossby wave signals are evident in OLR signals filtered to Kelvin and Rossby Wave frequencies and wavenumber.

The core findings of Matthews (2004) were reinforced by Maloney and Shaman (2008) using Tropical Rainfall Measurement Mission (TRMM) precipitation data for West Africa and a slightly different temporal filtering approach. They found a peak in rainfall variability on 50 day timescales and tested a number of potential mechanistic associations, including possible relationships between the rainfall anomalies and water vapor flux, boundary layer divergence, and isentropic potential vorticity anomalies that could lead to enhanced AEW activity. None of these analyses yielded significant or coherent associations. The only convincing associations were between WAM 50 day rainfall variability and eastward-propagating upper level anomalies (200 hPa zonal winds and 300 hPa temperature) originating in the Western Pacific warm pool, consistent with the Matthews (2004) dry Kelvin wave mechanism.

While Maloney and Shaman (2008) emphasized the role of dry Kelvin waves, other observation-based studies found a greater contribution of westward-propagating Rossby waves. Using OLR diagnostics and a dry primitive equation atmospheric model, Janicot et al. (2009) show that 25-90 day bandpassed OLR anomalies in the WAM region are associated with MJO wave activity, but they emphasize that these anomalies show a westward propagating tendency and are more strongly associated with convectively coupled Rossby waves; they find the dry Kelvin wave contribution to be relatively weak. At the same time, Janicot et al. (2009) emphasize that WAM variability on this timescale is not solely driven by MJO wave activity, and that a teleconnection between the Indian monsoon and WAM is also significant. Janicot et al. (2010) further this line of inquiry, confirming the importance of MJO-associated Rossby waves to WAM variability on 30-100 day timescales. They emphasize that this variability is associated with intensification and reduction of ITCZ convection and not ITCZ displacement, which is important for shorter

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timescale WAM variability. Pohl et al. (2009) also stress the importance of Rossby waves, and they note that the MJO Rossby wave impact is much more significant for dry spells than for wet spells—i.e., Rossby wave induced suppression of WAM convection is the dominant influence. This finding is particularly relevant for predictive applications, as dry spells within the WAM are of considerable importance for agriculture (Gadgil and Rao 2000; Sultan et al. 2005).

The studies described above were primarily diagnostic, with some use of simplified physically-based models. This approach has the advantage of empiricism but has limited power to distinguish cause and effect in complex interacting systems. For this reason, recent studies have incorporated global climate model (GCM) experiments designed to isolate MJO impacts on the WAM. Lavender and Matthews (2009) and Mohino et al. (2012) each performed controlled GCM experiments to determine whether WAM variability could be forced as a response to the MJO. Lavender and Matthews (2009) forced an MJO-like signal on an atmosphere-only version of the Met Office Unified Model GCM by imposing observed MJO SST anomalies in the equatorial warm pool. Mohino et al. (2012) adopt a different approach, forcing MJO features into the LMDZ atmospheric GCM by nudging atmospheric fields in the equatorial warm pool region with observations from the ERA-40 and ERA Interim reanalyses. Both approaches produce a semblance of MJO dynamics that can then be compared to control model simulations that do not include realistic MJO activity. The two studies find that imposing an MJO on a model that does not otherwise include one does produce a rainfall response in the WAM region. In both cases, the influence of convectively coupled, westward-propagating Rossby waves was found to be a robust mechanistic link between the MJO convective zone and the WAM. The modeling results from Mohino et al. (2012) suggest that the Rossby wave mechanism is almost exclusively responsible for the association, as dry Kelvin waves in their model did not propagate far enough to influence the WAM. Lavender and Matthews (2009) also find Rossby waves to be of primary importance, but their model simulations suggest that dry Kelvin waves also play a role.

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Both Lavender and Matthews (2009) and Mohino et al. (2012) had to impose MJO activity on GCMs that otherwise failed to produce a realistic MJO signal. The failure of those GCMs to produce an MJO on their own was not surprising, given the lack of MJO dynamics in most GCMs up through and including the 3rd Coupled Model Intercomparison Project (CMIP3) generation of models (Slingo et al. 1996; Waliser et al. 2003a; Zhang et al. 2006; Lin et al. 2006; Kim et al. 2009; Waliser et al. 2008). In an encouraging development, however, studies by Hung et al. (2013) and Niang et al. (2016) have found that CMIP5 models show significant improvement in the simulation of the MJO. Niang et al. (2016) specifically examined the representation of MJO impacts on WAM in the atmosphere-only (AMIP) experiments performed under CMIP5. They find that most models are capable of simulating an MJO and of capturing the association between MJO activity and WAM rainfall. Interestingly, they find that in these models eastward propagation of the MJO core has the largest impact on WAM, followed by westward propagating Rossby waves. Similar to Mohino et al. (2012), dry Kelvin waves tended to die out before reaching the WAM zone. The ability of CMIP5 models to capture MJO influence on the WAM has implications for dynamically-based forecast systems, as will be addressed in Section 6.

*3.1.2 The role of African Easterly Waves*

A complementary set of studies of summertime MJO influence on West Africa has focused on the potential for MJO to influence the initiation of African Easterly Waves (AEW), which are baroclinic disturbances with a wavelength of 2000-5000 km and a period of 3-5 days (Reed et al. 1977; Burpee 1972). AEW are, in turn, understood to alter convection patterns in West Africa and the tropical Atlantic Ocean (Janiga and Thorncroft 2016).

The line of research that associates MJO with AEW variability builds from the observation that AEW activity varies on sub-seasonal timescales in boreal summer. Leroux et al. (2010) diagnosed variations in AEW activity in the NCEP2 reanalysis on the basis of high frequency (<

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6 day) variability in perturbation kinetic energy (PKE) in the 700 hPa and 850 hPa wind fields. They then passed high frequency PKE fields through a 10-120 day filter in order to identify synoptic drivers of PKE activity. This analysis yielded coherent intra-seasonal variability in AEW activity on 10-120 day timescales.

Leroux et al. (2010) also found that enhanced AEW activity was preceded by periods of strong convection in the AEW generation zone, located in the eastern Sahel and western edge of the Ethiopian highlands (see Figure 1). The presence of enhanced convection in the AEW generation zone leading AEW activity is consistent with the theory of AEW generation proposed by Thorncroft et al. (2008). It also offers a potential mechanistic link between the MJO and West African rainfall via AEW: the MJO enhances (or suppresses) convection in the AEW generation zone, leading to enhanced (suppressed) AEW activity and increased (decreased) rainfall in West Africa. This does not prove that MJO is involved in AEW variability, nor that it is the dominant driver even if it is involved. But it does suggest that some component of the MJO influence on West African summer rainfall could be explained through analysis of the MJO impact on convection in the upstream AEW generation zone.

Noting the results of Leroux et al. (2010), Ventrice et al. (2011) directly investigate the role of MJO in AEW variability. Using the RMM phases as a basis for analysis, they diagnose associations between MJO phase and PKE in West Africa. They find significant and coherent variability in PKE associated with MJO phases, which they interpret as enhanced AEW activity when the MJO convective center is located over Africa or in the Indian Ocean (RMM phases 1-3). Further, they find that enhanced OLR in the AEW generation region and African Easterly Jet (AEJ) activity are both associated with the MJO with a one phase lead (phases 8, 1, and 2), suggesting that MJO can influence AEW generation both by influencing convection in the AEW generation region (i.e., the Thorncroft et al. (2008) mechanism) and by altering AEJ dynamics in a manner favorable to AEW. Ventrice et al. (2011) find evidence that the mechanistic link

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between MJO and these processes is Rossby wave activity, which is consistent with the studies of Matthews (2004), Janicot et al. (2009), Pohl et al. (2009) and others. Notably, however, the AEW-based analysis associates Rossby wave activity with convection in the AEW generation zone, which in turn influences AEWs that propagate into the WAM region. This is distinct from the hypothesized direct impact of Rossby waves on WAM convection.

This distinction is clear in an MJO-AEW study performed by Alaka and Maloney (2012), which focuses on MJO influence in the AEW generation zone. Using satellite-derived brightness temperature data and ERA Interim reanalysis, they find that convection anomalies and PKE in the AEW generation zone are systematically associated with MJO phases, and that these anomalies precede WAM convection anomalies by ~10 days. This result is consistent with the Thorncroft et al. (2008) hypothesis that convection in the AEW generation zone influence AEW formation, and it also suggests that WAM convection anomalies on MJO timescales are a downstream product of MJO influence in the AEW generation zone. Interestingly, since the AEW generation region is located to the east of the WAM, MJO-generated Rossby and Kelvin waves do not arrive synchronously—Rossby waves lead Kelvin waves by 5-10 days. This makes it possible to diagnose the impact of each wave separately without the complication of deconvolving overlapping signals. Alaka and Maloney (2012) find that both Kelvin and Rossby waves influence AEW activity, though the mechanisms of influence may differ between the two wave types.

There is a broad consistency between studies of the direct MJO influence on WAM convection and the studies that focus on MJO influence on AEW. Both have identified robust relationships between MJO and intra-seasonal WAM rainfall variability, and both find that some combination of Rossby and Kelvin waves are the basis for the association. But there are also interesting contrasts. For example, Lavender and Matthews (2009) concluded that AEW activity on MJO timescales is a product of WAM convection rather than a precursor, while Alaka and Maloney

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(2012) suggest that the MJO influence on WAM convection is a downstream result of processes in the AEW generation zone. These findings are not necessarily mutually exclusive, but they do point to complications in the MJO-WAM connection that are yet to be resolved.

*3.1.3 Springtime West African Rainfall*

While the majority of West African rainfall studies have focused on summer, springtime rainfall (March – June) is critically important in the southern portion of the region, including the Gulf of Guinea coast and Equatorial West Africa. In this area spring rains allow for an important cropping season, and they can account for up to 45% of annual average rainfall (Eugène et al. 2012; Berhane et al. 2015). There is also reason to expect that MJO influence on precipitation in this season could be quite significant, as it coincides with months of strong MJO activity.

Indeed, Gu (2009) applied a diverse set of satellite-derived data to assess MJO associations with intra-seasonal precipitation variability in the West African spring rains and found a significant signal. He notes that the MJO-associated convective variability in West Africa appears to be an extension of an eastward propagating signal that affects the entire Tropical Atlantic basin, and he further proposes that the convective signal in West Africa could be strong enough to contribute to MJO propagation. Laing et al. (2011) also note the potential for MJO events to influence rainfall in Equatorial Africa (including Equatorial West Africa) and identify a number of specific events in which the eastward-propagating MJO convective core passes through Equatorial Africa, with strong influences on precipitation.

Berhane et al. (2015) also emphasize the importance of the MJO convective core during the spring rainy season. Similar to several studies of MJO during WAM, they apply wavenumber-frequency analysis to isolate the relative contributions of dry Kelvin waves, Rossby waves, and the MJO convective core, and they find that the convective core is dominant. Rossby and dry Kelvin signals are also evident, suggesting that all three mechanisms of communication are active in this season. Berhane et al. (2015) compared their spring season results to an identical

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analysis performed for summer rainfall. They find that the MJO influence is greater for the spring rains, with MJO-associated daily rainfall variability on the order of 20-50% of average daily rainfall during that season. This influence is strongest in March and declines as the season progresses. Interestingly, Berhane et al. (2015) find that the influence of the convectively coupled MJO core is significantly stronger in spring than in summer (Figure 5a,b), but the Rossby wave influence is similar for the two seasons (Figure 5c,d). This difference could be related to the influence that Indian Monsoon interactions with MJO have on wave propagation during the summer season. They also find a small but distinct Kelvin Wave signal in spring but not in summer (Figure 5e,f).

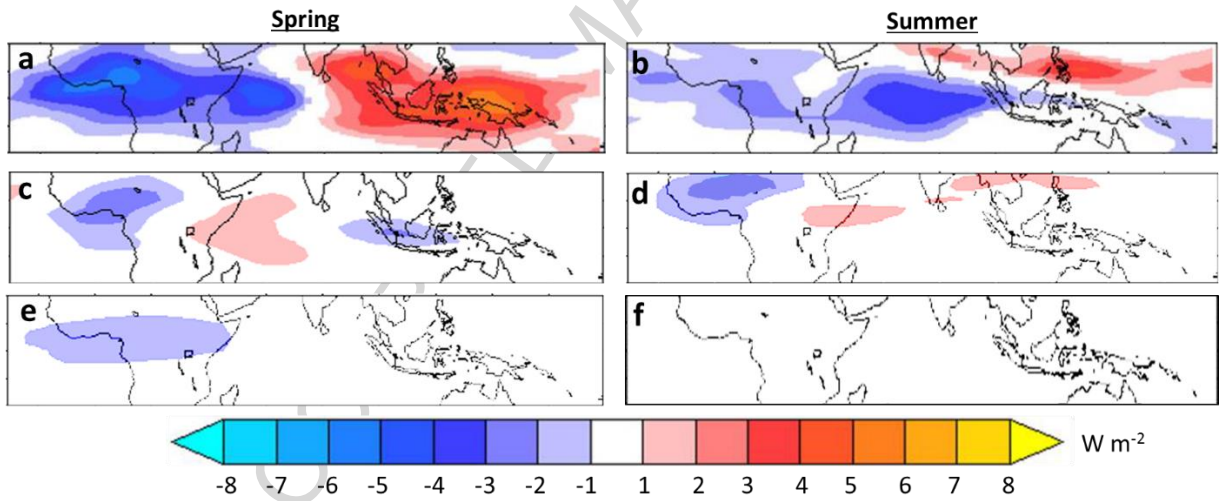


*Figure 5: Regression of zero-lag OLR against a customized MJO index that represents active phase MJO over West Africa, for the period 1980-2013. The OLR signal is isolated for (a,b) MJO, (c,d) Rossby wave, and (e,f) Kelvin wave through wavenumber-frequency filtering. Left column (a,c,e) is for spring (March-June) and right column (d,b,f) is summer (July-September). (Figure adapted from Berhane et al. (2015); methodological details provided in that paper.)*

## 3.2 East Africa

Compared to West Africa, relatively fewer studies have been performed on MJO influence in East Africa. These studies have focused almost exclusively on equatorial East Africa (EEA; Figure 1), a region that experiences two rainy season: the "long" rains (March – May) and the

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"short" rains (October – December). The relative amount of rainfall in these two seasons differs by location, but both are important for agriculture and water resources. Notably, the two seasons show different sensitivity to large scale drivers (Camberlin et al. 2009), and the character of precipitation changes over the course of each season, as the ITCZ passes across the region (Camberlin and Okoola 2003). In contrast to West Africa, EEA is located relatively close to active MJO convective regions in the Indian Ocean and, to some extent, the Maritime Continent and Western Pacific. This leads do a different family of mechanistic links between MJO and local rainfall, as MJO convection in the Indian Ocean or Western Pacific can influence EEA through synoptic scale circulations even in the absence of long range wave propagation.

Mutai and Ward (2000) first noted that a mode of intra-seasonal rainfall variability in the EEA short rains matches MJO timescales. In a diagnostic study performed using Climate Research Unit (CRU) station-based rainfall data for EEA combined with satellite-derived OLR and NCEP/NCAR Reanalysis data, they found that enhanced MJO convection in the Indian Ocean lags positive rainfall anomalies in EEA by approximately 5 days. As this is a time lagged association, Mutai and Ward (2000) suggest that EEA convection might contribute to the formation of MJO events in the Indian Ocean. They do not investigate mechanisms through which MJO features prime EEA rainfall, but they do note the potential importance of scale interactions between intra-seasonal oscillations, including the MJO, and inter-annual climate variability affecting EEA rainfall. (Pohl and Camberlin 2006a). (Pohl and Camberlin 2006b)

Later studies, however, did identify mechanisms through which MJO drives intra-seasonal EEA rainfall variability. In a diagnostic study that combined weather station data from Kenya and Tanzania with NCEP/NCAR Reanalysis data and satellite-derived OLR, Pohl and Camberlin (2006a) found MJO associations with EEA rainfall in both the short and long rains. For both seasons they find a robust link between deep convection over the EEA highlands and the eastward-propagating MJO center of convection. Notably, the EEA convection anomalies in

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their analysis lag the MJO pass through EEA, and they are strongest when MJO convection is in the Indian Ocean. Pohl and Camberlin (2006a) interpret this lag as a result of the time required for convection to develop in EEA after MJO-induced destabilization and/or as an indication that EEA convection results from westerly wind intrusions, which are pulled into the region when the MJO low pressure center is located in the western Indian Ocean. They also note that the mechanism of association between MJO and highland rainfall in EEA is distinct from the association between MJO and rainfall in the coastal lowlands. Coastal rainfall anomalies do show significant association with MJO, but these associations are out of phase relative to the highlands, with enhanced coastal rainfall occurring prior to MJO arrival in EEA. This could indicate that coastal rainfall is enhanced when MJO-associated low pressure in the Atlantic sector leads to strengthened easterly trade winds, which bring moist air from the Indian Ocean into coastal EEA. Pohl and Camberlin (2006a) further note that this coastal rainfall anomaly is stratiform in nature, and does not show large OLR anomalies that would be associated with deep convection.

Omeny et al. (2008) investigate statistical associations between MJO and Kenyan rainfall from an operational perspective. Similar to Pohl and Camberlin (2006a) they find significant correlation between highland rainfall—in this case western Kenya rainfall—and MJO when the MJO center of convection is in the Indian Ocean. Reduced rainfall in western Kenya is found when MJO moves into the Western Pacific. The result holds for both short and long rains. They note that this association could be used to inform long-range rainfall forecasts on the order of 10 days, but they recommend combining MJO information with other diagnostics in order to explain a larger fraction of variability. Omeny et al. (2008) find no significant associations between MJO and rainfall in eastern Kenya.

Berhane and Zaitchik (2014) add detail to the studies of Pohl and Camberlin (2006a) and Omeny at al. (2008) by considering sub-seasonal variability in the MJO influence on EEA within

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both the short and long rainy seasons. Using satellite-derived rainfall and OLR data in combination with multiple reanalysis datasets, they find that MJO convection in the Indian Ocean is associated with enhanced EEA highland rainfall throughout the long rains, but that the association is strongest at the end of the season (May). During the short rains, significant associations with highland rainfall are relatively week in October but are strong in November and December (Figure 6a-c). In October there is, however, a strong negative association between Maritime Continent MJO convection and coastal EEA rainfall that is not found later in the season (Figure 6d-f). From a prediction perspective, these associations are significant at up to four pentad lead for specific months.

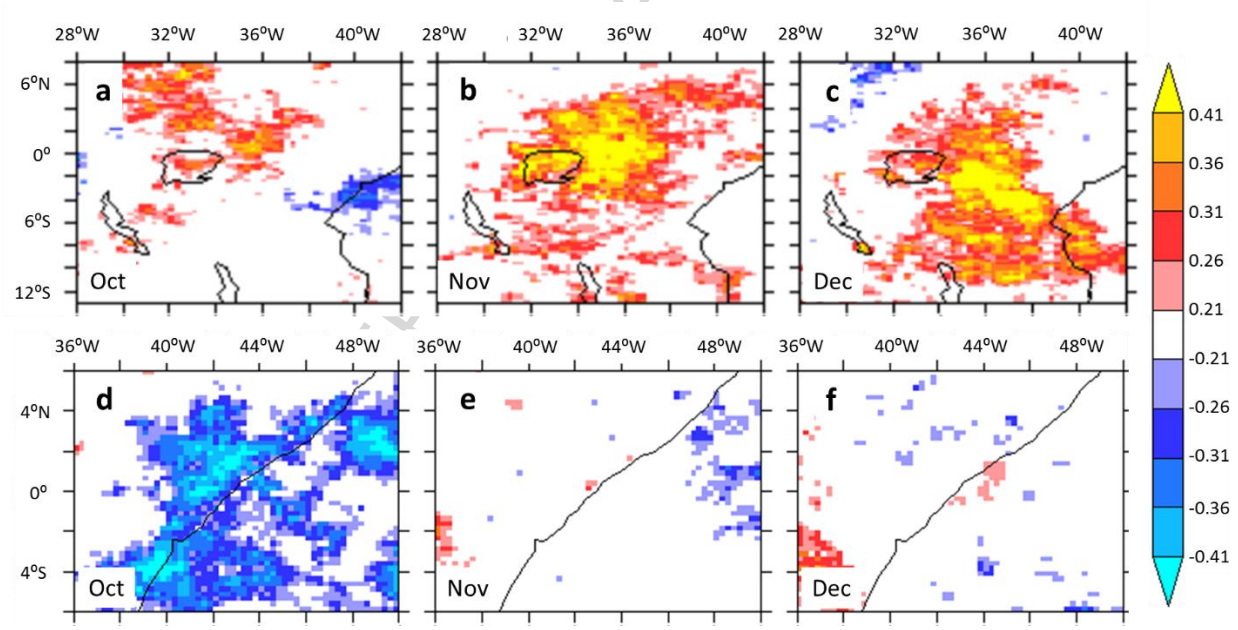


*Figure 6: Correlation between daily precipitation (CHIRPS) and MJO convective activity in the (a-c) Indian Ocean, for MJO convection centered on 80 $^{0}$E; (d-f) Maritime Continent, for MJO convection centered on 120 $^{0}$E, during the October-December short rains. NOAA OLR MJO index was used in this analysis. Details provided in Berhane and Zaitchik (2014).*

Berhane and Zaitchik (2014) also provide analysis of the mechanisms of MJO influence on the region. For the highlands, they find that the MJO influence is communicated through modification of the low-level southeasterly winds that enter EEA from the Indian Ocean. This modification influences both temperature advection, influencing atmospheric stability, and low-

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level convergence patterns. In the coastal region, they find that MJO convection over the Maritime Continent in October strengthens the Indian Ocean branch of the Walker circulation. This leads to stronger deep atmospheric subsidence over coastal EEA as well as a stronger Somali Low Level Jet, which creates shallow frictionally-induced coastal subsidence. Together, these mechanisms suppress coastal EEA rainfall when MJO is located in the Maritime Continent and Western Pacific.

The question of the strength of MJO association with EEA rainfall that Omeny et al. (2008) raise in the operational context is also examined by Pohl and Camberlin (2006b). Using similar data as applied in Pohl and Camberlin (2006a), the authors examine inter-annual variability in the strength of MJO influence on EEA during the long rains. They find that the correlation between smoothed EEA highland rainfall and MJO indices varies from 5% to 53% from year to year, which they attribute to variability to the character of the MJO. This character can vary from event to event as well as within a season or from year to year.

Intriguingly, at inter-annual timescales, Pohl and Camberlin (2006b) find that 44% of inter-annual variability in the long rains can be explained by an index of MJO amplitude: years of stronger MJO activity are associated with an earlier onset of rainfall and a larger number of heavy rainfall periods, resulting in higher seasonal total rainfall. They find that the relationship between MJO amplitude and EEA rainfall is partially mediated by ENSO. While efforts have been undertaken to study the relationship between intra-seasonal and inter-annual rainfall variability in EEA using GCMs (Philippon et al. 2015), no study to date has investigated the proposed ENSO-MJO-EEA relationship with a dynamically-based model.

### 3.3 Southern Africa

Austral summer is the primary rainy season in subtropical Southern Africa, with most rain falling between October and April. This coincides with seasons of strong MJO activity, suggesting that MJO has the potential to influence precipitation variability in this region.

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However, relatively few studies have been performed on the relationship between MJO and rainfall in Southern Africa, and many questions remain about the strength and mechanism of MJO influence.

The study that focuses most specifically on MJO and Southern African rainfall is Pohl et al. (2007). Their analysis indicates that MJO events account for 35-40% of intra-seasonal OLR variance in the region. This variance is primarily associated with the migration of the MJO convective core and is out of phase with tropical Africa: there is no coherent OLR signal in Southern Africa when MJO convection is maximum in tropical Africa, but significant negative OLR anomalies (enhanced convection) appear when tropical Africa is in an MJO suppressed phase (Figure 7). Pohl et al. (2007) also find that anomalies in rain gauge data on MJO timescales are associated with systematic large scale atmospheric anomalies: during wet periods there is widespread northerly flow in the region and strengthening of 700 hPa easterly winds over the Congo Basin. In addition, they find that MJO is associated with a convection dipole between Southern Africa and the Southwest Indian Ocean that is known to have significant impacts on intra-seasonal rainfall variability (Jury 1992). However, Pohl et al. (2007) note that MJO is not the only influence on this dipole and is only one of many factors influencing rainfall variability in the region. In particular, they identify tropical-temperate troughs (TTT) as a significant source of variability and of tropical influence that could be related to MJO.

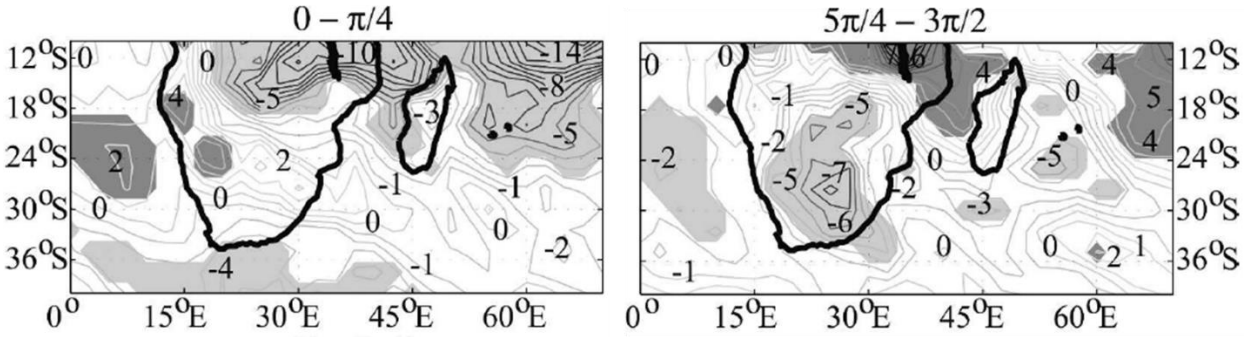


*Figure 7: Composites of deseasonalized OLR anomalies (W m$^{-2}$) based on the MJO indexing scheme of Pohl et al. (2007). The left panel shows anomalies for the phase of active MJO convection over tropical Africa, as evident in the OLR signals at the north of the map. The right*

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*panel is for MJO suppressed phase over tropical Africa. Shading indicates significance at 95%. Figure is from Pohl et al. (2007) and is reprinted here with permission of the American Meteorological Society.*

Subsequent studies of MJO mechanisms of action on Southern Africa have yielded additional insights, but results are often ambiguous. Pohl et al. (2009) found that TTT do show variability on MJO timescales, but that they are equally likely to occur during any MJO phase and are thus an independent mode of variability. Hart et al. (2012) investigate MJO-TTT relationships in southern Africa using a novel object detection technique to diagnose TTT from satellite-derived OLR. They find a significant but weak association, with TTT formation suppressed when the MJO center of convection is located over tropical Africa and enhanced when MJO convection is in the Western Pacific. They suggest that this association could be related to moisture flux variability into subtropical Africa, but note that a longer time record is required to confirm the relationship. Oettli et al. (2014) introduce the use of self-organizing maps (SOM) as a means for identifying nonlinear associations between MJO and Southern Africa. They find that SOM identify MJO relationships with both the Southern Africa - Southwest Indian Ocean convection dipole and, to some extent, TTT activity, but the phase of their TTT results is inconsistent with Hart et al. (2012). More recent studies have suggested that MJO modulates rainfall yield of TTT in Southern Africa rather than initiation of TTT, which would explain why MJO phase relationships with TTT occurrence are weak even in the presence of an MJO influence (Oettli et al. 2014; Hart et al. 2013). No targeted dynamical modeling experiments have been performed to investigate the relationship between MJO and TTT in this region, however, and further work is clearly required.

## 4. Mechanisms of MJO Influence on Africa

The mechanisms of MJO influence on Africa fall into three categories: (1) direct influence of the MJO convective core, as it passes over Africa; (2) synoptic influence of the convective core when

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it is located in or near Africa; (3) remote teleconnections communicated via dry Kelvin and/or Rossby waves emanating from the MJO convective region. Not surprisingly, as reviewed in Section 3, relative emphasis on each of these classes of influence differs by region. This is largely a function of geography: the MJO is most active in the Indian Ocean and Western Pacific, so the further a region is from that area (e.g., West Africa) the more likely it is that MJO influence depends on remote teleconnections. But one must also consider the possibility that differences in emphasis simply reflect the interests of the few research groups that focus on the problem of MJO impacts on Africa. This is a small and evolving literature, and it is possible that new perspectives and analytic techniques will modify our understanding of which mechanisms are most important to each region within Africa.

Indeed, identification and characterization of MJO influence appears to be particularly sensitive to method of analysis. For example, prior to Matthews (2004), the majority of studies failed to identify a robust MJO signal in West Africa, as this signal only becomes clear with an adequate data record and selective temporal filtering, aided by a regionally customized MJO definition. In Southern Africa, Pohl et al. (2007) found that the impact of MJO was clear in composite plots but not in spectral analysis, perhaps because higher frequency modes of variability detract from the MJO signal. Studies with dynamical models are sensitive to the model's representation of MJO, the method used to impose or remove the MJO signal, and the strength of wave propagation within the model. Differences between models could be the reason that modeling studies of West Africa emphasize the importance of propagating Rossby waves (Lavender and Matthews 2009; Mohino et al. 2012) or the eastward propagation of the MJO convective core (Niang et al. 2016) to different degrees. Model limitations might also explain why the influence of dry Kelvin waves on West African rainfall is minimal in GCM-based studies but significant in most (though not all) diagnostic studies. Finally, as is noted in several of the studies reviewed for this paper, the use of discrete indices to define MJO phases is limiting. MJO propagation is a continuous process that is imperfectly captured by any of the categorical index systems. Choice

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of index system can influence the results of a diagnostic study, and it is also possible that some processes simply aren't captured effectively by any of the prevailing methods for classifying MJO phases.

Bearing those limitations in mind, it is still useful to assess the current state of mechanistic understanding for each region of Africa (Table 1). For the West African summer season, there is a clear emphasis on the influence of westward propagating equatorial Rossby waves, with additional dry Kelvin wave influence identified in some but not all analyses. The means through which the Rossby waves influence precipitation, however, is very much open to continued research. Studies that focus on the direct influence that Rossby waves have on WAM convection note the potential for these waves to alter atmospheric stability profiles in the WAM zone, but whether this is primarily due to changes in tropospheric temperature or to altered wind shear patterns associated with the AEJ is not clear. A parallel line of analysis holds that the primary Rossby wave influence is due to the impact these waves have in the upstream to the East, at the entrance to the AEJ. According to these studies, MJO Rossby waves alter conditions relevant to AEW generation in this region through their impacts on convection or on the AEJ, and it is the AEW that then influence convection in the WAM region. The relative importance of direct Rossby wave influence on the WAM convective environment versus indirect influence via AEW generation is a significant distinction. Improved understanding of these processes through dynamical models and enhanced diagnostic studies could help to explain and quantify the MJO influence on WAM, shed light on the relationship between AEW, WAM convection, and Atlantic tropical cyclone development, and inform dynamical model development to capture MJO influence appropriately.

*Table 1: Summary of the mechanisms of MJO influence that have been emphasized in studies of African precipitation, grouped by rainfall season and region. Underline indicates the mechanism that has been identified most consistently across studies. Note that mechanisms are not mutually exclusive. Months listed are approximate, as studies sometimes differ in their definitions of the rainy season.*

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| | West Africa (Jun-Oct) | West Africa (Mar-Jun) | East Africa (Oct-Dec) | East Africa (Mar-May) | Southern Africa (Nov-Apr) |
|---|---|---|---|---|---|
| **Convective core** | X | **X** | X | X | |
| **Indirect synoptic dynamics** | X | | **X** | **X** | **X** |
| **Rossby waves** | **X** | X | | | |
| **Kelvin waves** | X | X | | | |

Studies of Equatorial East Africa have not focused on the influence of Rossby or dry Kelvin waves. In this region the emphasis has been on the direct influence of the MJO convective core or on the synoptic effects of MJO convection in the neighboring Indian Ocean and more distant Maritime Continent and Western Pacific regions. The possibility of direct influence was noted by Pohl et al. (2006a): the MJO convective core destabilizes the atmosphere on its pass through the region, but it takes several days for this destabilization to yield enhanced convection. This leads to a time lagged association that is in fact the result of a direct influence of MJO convection. Alternatively, synoptic scale dynamics might explain the association. This was the emphasis of Berhane and Zaitchik (2014), who identified a dynamical link between MJO convection in the Indian Ocean and southeasterly winds entering the East African highlands. This leads to an MJO influence on convection due to thermodynamic and convergence effects of altered low level winds. For coastal East Africa, both Pohl et al. (2006a) and Berhane and Zaitchik (2014) focus on synoptic phenomena, but with different interpretations of mechanism: Pohl et al. (2006a) suggest that MJO influences the strength of trade winds entering East Africa, leading to changes in stratiform precipitation along the coast, while Berhane and Zaitchik (2014) identify a Walker Circulation influence, with enhanced Walker Circulation subsidence occurring on the East African coast when the MJO convective core is located in the Maritime Continent.

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Distinguishing these mechanisms will likely require modeling studies that have not yet been performed for this region.

Southern Africa stands out as the one primarily extratropical region considered in this review. Studies of Southern Africa have focused almost exclusively on the region south of $15^0$S, covering the Southern Africa subtropics and on to the edge of mid-latitude dynamics. The mechanism and even the magnitude of MJO influence on this region is still poorly characterized. Leading hypotheses focus on synoptic effects of the MJO convective core. One possibility is that the MJO core influences the intensity of rainfall associated with TTT as they pass through Africa, producing convective anomalies in Southern Africa that are out of phase with anomalies in equatorial Africa (Oettli et al. 2014; Hart et al. 2013). Another possibility is that MJO acts through its influence on a convection dipole between the Southwest Indian Ocean and Southern Africa. MJO could influence that dipole through its impacts on convection when the convective core is located in the Indian Ocean sector. Studies to date have yielded mixed results on both of these proposed mechanisms. Given the complexity of precipitation processes in this region, including multiple drivers of intra-seasonal variability on MJO-like timescales, it would seem that controlled modeling experiments will be needed to disentangle the MJO mechanism of influence. Better *in situ* precipitation records from subtropical and tropical Southern Africa would also be valuable, as gauge-based studies of the MJO influence on precipitation have been restricted to the southern edge of the region (South Africa) or have had to make use of limited and questionable data records from further north (Pohl et al. 2007).

## 5. Influence of ENSO and Other Climate Modes

As noted in a number of studies, and emphasized by Pohl and Camberlin (2006b), the MJO influence on African precipitation is inconsistent. Even in regions that show significant associations over an extended period of analysis, the importance of MJO relative to other drivers of variability and the strength of the MJO-rainfall association vary significantly from year to

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year, and even from event to event. This presents an important challenge to research and forecast applications. Even a statistically robust relationship between MJO and rainfall is unsatisfying and ultimately of questionable forecast value if there is unexplained and unpredictable variability in the strength of the association. One promising hypothesis is that ENSO and other relatively slowly varying modes of climate variability influence the relationship between MJO and African rainfall on inter-annual time scales, and that accounting for the state of these climate modes can at least provide a probabilistic prediction of how individual MJO events will influence rainfall within a season.

The potential for slowly varying climate modes to influence MJO impacts on Africa can be thought of as a two pronged problem. First, does the slowly varying mode alter MJO frequency, strength or propagation? This is relevant when considering the relative importance of MJO for sub-seasonal variability in any given year, and it matters for event-scale prediction in cases where the mechanistic link depends on propagation of the MJO convective core through the region of interest. Second, does the slowly varying mode influence the strength of MJO teleconnections to other regions? This could be the case if ENSO, for example, alters background atmospheric or oceanic conditions in a way that changes dry Kelvin and Rossby wave propagation or synoptic flow patterns that communicate MJO influence beyond the convective core.

The large majority of literature on MJO interactions with slowly varying climate modes has focused on ENSO. The ENSO literature can be further subdivided into studies concerned with the impact that MJO has on ENSO and studies that examine ENSO impacts on the MJO. In the first category, a substantial number of studies have found that strong MJO convective events are associated with westerly wind bursts known to be important to the development of El Niño conditions (Harrison and Schopf 1984; Kessler et al. 1995; Belamari et al. 2003; Wang et al. 2013; Seiki and Takayabu 2007a; Seiki and Takayabu 2007b; Kapur and Zhang 2012; Zhang

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2005), while others have noted the potential for nonlinear feedbacks between MJO and ENSO (Roundy and Kravitz 2009). This would mean that MJO intra-seasonal variability has implications for seasonal and inter-annual climate variability via its impacts on ENSO. While this line of inquiry has implications for African climate variability, it is not directly relevant to the question of how the MJO itself influences African rainfall.

The second, more directly relevant line of MJO-ENSO research concerns the impact that ENSO state has on the character of MJO events. While there is no obvious synchronous correlation between ENSO and inter-annual variability in MJO activity (Slingo et al. 1999; Hendon et al. 1999), relationships do exist when one considers time lagged associations, metrics of MJO location and propagation, and the character of each ENSO event. Time lagged associations include the observation that MJO is particularly strong in the Pacific Ocean as the warm tongue of an El Niño event develops, and that it is particularly weak in the Pacific in the wake of an El Niño event (Zhang and Gottschalck 2002). Spatially, MJO activity tends to shift eastward (westward) during El Niño (La Niña) events (Anyamba and Weare 1995; Fink and Speth 1997; Gualdi et al. 1999; Hendon et al. 1999; Woolnough et al. 2000; Bergman et al. 2001). Impacts on propagation speed are more complicated. Tam and Lau (2005) find that El Niño slows MJO propagation in the Pacific Ocean, likely due to the impact of altered SST gradients, while Pohl and Matthews (2007) find that it speeds propagation in the Maritime Continent and into the Western Pacific, possibly because reduced convection during El Niño allows for faster propagation of convectively coupled Kelvin waves. Indeed, El Niño events have been found to be associated with weaker MJO convection in the equatorial Pacific (Chen et al. 2016). The two results are not necessarily inconsistent, as the focus region was different, but Pohl and Matthews (2007) note that their finding of faster propagation during El Niño means that MJO lifecycles are shorter in El Niño even though the propagation distance into the Pacific is further. ENSO could also influence MJO activity on longer timescales. Pohl and Matthews (2007) use

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reconstructed MJO indices to find that the MJO was likely weaker prior to the ENSO break of 1975 and has been stronger, on average, in the years since.

Several recent studies have focused on the importance of ENSO character (or "flavor") to ENSO-MJO interactions. These studies distinguish between the Eastern Pacific (EP) and Central Pacific (CP) or "Medoki" types of El Niño (Yu and Kao 2007; Ashok et al. 2007). Gushchina and DeWitt (2012) found that enhanced MJO activity tends to precede the mature phase of EP El Niño events, perhaps contributing to their formation, while it is active primarily in the mature and decaying phases of CP El Niños, suggesting that it is more important to persistence of these events than to triggering them. In contrast, Feng et al. (2015) find that MJO is active in the early development stages of CP El Niños, and could serve to trigger these events. Their results for EP El Niños show that mature EP El Niño conditions suppress MJO convection in the Maritime Continent, and they do not find evidence of MJO as an EP El Niño trigger. Chen et al. (2016) arrive at results that are more similar to Gushchina and DeWitt (2012) than to Feng et al. (2015), but further emphasize that both types of event can suppress MJO activity over the Maritime Continent, with EP providing more effective suppression because of stronger atmospheric subsidence. They find that MJO propagates further into the Pacific Ocean during CP events on account of moistening of the lower atmosphere eastward of the MJO convective core. All of these studies are, however, limited by the small number of events available for analysis.

The second question posed in this section—whether ENSO or other modes alter atmospheric or oceanic states in a way that affects MJO teleconnections—has not been studied as extensively. Roundy et al. (2010) point out that ENSO and MJO are both useful for long-range weather prediction around the globe, but that the two interact in nonlinear ways. For this reason, they conclude that composite analyses based on combined ENSO and MJO conditions are more useful for prediction than are analyses based on parallel evaluation of the two states. Their

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analysis emphasizes mid and high latitude MJO teleconnections, which depend on wave propagation dynamics that are modified by ENSO's impact on atmospheric conditions. Moon et al. (2011) also examine ENSO impacts on MJO teleconnections, again focusing on extra-tropical phenomena. They find that ENSO shifts the spatial pattern of extra-tropical MJO teleconnections in a manner that can modify and even reverse the weather impacts of an MJO event. Hoell et al. (2014) study the implications of MJO for ENSO air-sea coupling in the tropical Pacific. They find that MJO convection disrupts air-sea coupling and thus weakens the atmospheric branch of ENSO anomalies. This results in reduced ENSO-related rainfall anomalies in the tropical Pacific Ocean and Maritime Continent and also in dampened extra-tropical ENSO impacts across the Northern Hemisphere. Hoell et al. (2014) do not specifically examine teleconnections to Africa, but a weakening of ENSO atmospheric anomalies under active MJO conditions suggests that MJO could act to limit the magnitude of ENSO impacts on rainfall in Africa. However, in a study of South America, Shimizo and Ambrizzi (2015) find that the influence of MJO on ENSO teleconnections can vary regionally and with MJO phase. Roundy et al. (2010), Moon et al. (2011) and Hoell et al. (2014) all considered boreal wintertime conditions, when ENSO tends to be strongest. Shimizo and Ambrizzi (2015) consider both boreal summer and boreal winter and find that MJO modification of ENSO is greatest in boreal winter.

Beyond ENSO, researchers have also investigated the impact that the Indian Ocean Dipole (IOD; Saji et al. 1999) and the Quasi-Biennial Oscillation (QBO; Baldwin et al. 2001) have on MJO activity. Studies of IOD have consistently found that negative (positive) IOD events are associated with enhanced (suppressed) MJO convection in the Indian Ocean. This is true in observational analysis (Shinoda and Han 2005; Kug et al. 2009; Wilson et al. 2013) and features of the relationship can be produced with GCMs (Benedict et al. 2015). The negative phase of IOD is defined by warmer than usual eastern Indian Ocean SSTs relative to western Indian Ocean SSTs, which leads to humidity convergence in the eastern Indian Ocean that could

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promote the initiation of MJO events (Wilson et al. 2013). Both the enhancement of MJO activity and the eastward shift of convection during negative IOD events could be relevant to the MJO influence on Africa, but no study has addressed this topic.

The QBO describes variability in zonal winds in the tropical stratosphere. A positive QBO is defined by westerly winds and a negative QBO is defined by easterly winds. Yoo and Son (2016) find a robust observation-based relationship between QBO and MJO activity, in which MJO is more active during negative (easterly) QBO events. They hypothesize that positive QBO events can inhibit MJO development due to increased wind shear in the lower stratosphere and upper troposphere, which lowers the tropical tropopause and increases thermal stratification across the tropopause. This can reduce deep convection in the tropics (Nie and Sobel 2015), possibly including MJO convection. The QBO-MJO link has potential applications to MJO forecasts (Yoo and Son 2016), but no study has examined how QBO-MJO interactions might influence MJO impacts on Africa.

All studies cited so far in this section address general aspects of MJO dynamics and teleconnections, and are not specific to Africa. The only studies that do specifically consider the influence that ENSO modulation of MJO might have on African rainfall have focused on East African rainfall. Pohl and Camberlin (2006b) find that although the statistical association between ENSO state and MJO activity is weak, there is a tendency for weak MJO activity in spring (March – May) in the wake of wintertime El Niño events. These weak MJO seasons are, in turn, associated with reduced long rainy season precipitation in East Africa, suggesting that MJO could be a key mechanism through which El Niño influences East Africa. Pohl and Matthews (2007) further investigate the MJO-ENSO relationship using a simplified MJO definition to extend the analysis back to 1950. They find a strong seasonality to the MJO-ENSO association, with maximum MJO activity found under El Niño in boreal winter and under La Niña in boreal summer. They did not specifically test associations with African rainfall, but the

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presence of a statistically robust and seasonally specific association between MJO activity and ENSO has clear relevance to the problem.

Complementing these studies, Berhane (2016) performs an analysis of the East African short rains. He finds that the MJO influence on East Africa in this season is substantially modified by ENSO state: during an El Niño, RMM Phase 2 (Indian Ocean convection) is associated with strong positive rainfall anomalies in East Africa (Figure 8a), but during a La Niña the association is weak and spatially mixed (Figure 8b). For ENSO neutral conditions there is some positive association, but it is not nearly as strong as for El Niño conditions (Figure 8c). Notably, these patterns are reinforced by the fact that MJO activity is stronger in the Indian Ocean during El Niño and in the Maritime Continent and Western Pacific during La Niña. Together, the increased frequency of Indian Ocean MJO during El Niño and the stronger connection to East Africa during Indian Ocean active phase could explain a significant portion of the long known connection between ENSO and inter-annual variability in the short rains. As shown in Figure 8d and 8e, the association between ENSO and East African rainfall under non-MJO conditions is weak, with El Niño (La Niña) showing only a small tendency towards enhanced (reduced) rainfall relative to average conditions. Potential mechanisms for the combined ENSO-MJO impact on East African short rains include the influence that ENSO has on the Indian Ocean branch of the Walker Circulation and differences in moisture advection across the Indian Ocean in El Niño relative to La Niña events.

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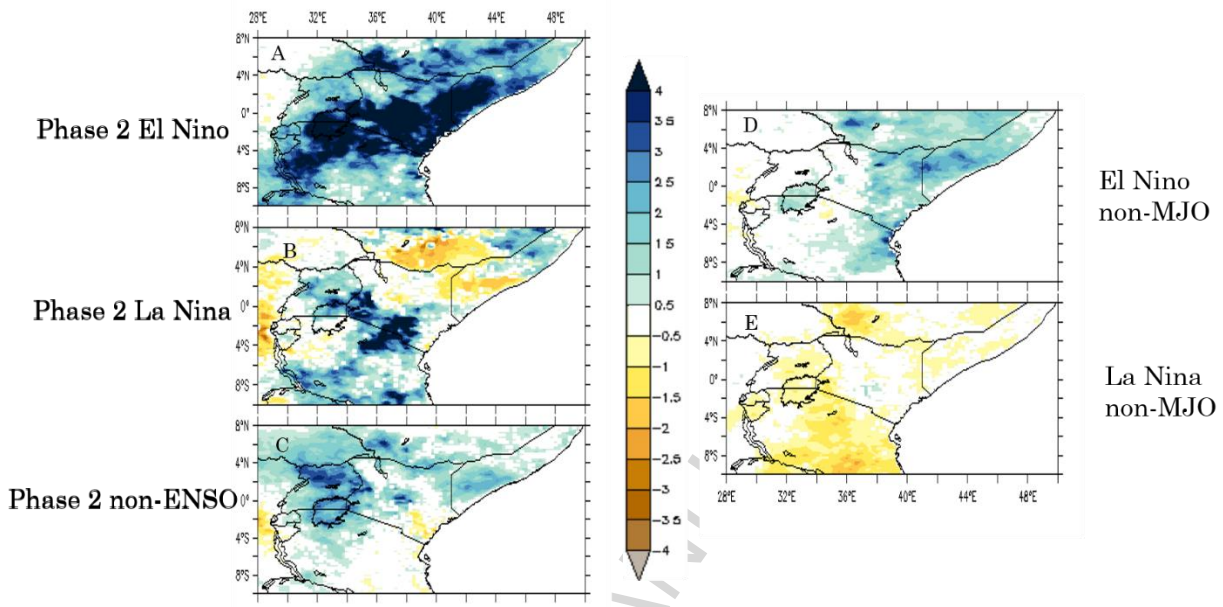


*Figure 8: Composites of East Africa short rain CHIRPS rainfall (mm/day). (A) RMM Phase 2 MJO and El Niño, (B) RMM Phase 2 MJO and La Niña, (c) RMM phase 2 MJO and non-ENSO, (D) El Niño and non-MJO and (E) La Niña and non-MJO. Figure from Berhane (2016) and reproduced with permission from the author.*

These studies of ENSO modulation of the MJO influence on African rainfall are limited, but they touch on a potentially important possibility: that MJO is an important mechanism for ENSO influence on East Africa. This finding could potentially inform S2S forecasts and longer term climate projections, if the level of MJO activity can be predicted or projected on these timescales. However, as explained in the following section, there is significant uncertainty in models of the MJO on these longer time horizons.

## 6. Forecasts and Projections

Sub-seasonal to seasonal (S2S) forecast is both a grand scientific challenge for Earth Systems Models and a potentially high value target for decision support. These dual motivations have inspired a number of initiatives to improve model performance on S2S timescales, including the World Climate Research Programme (WCRP) and World Weather Research Programme (WWRP) initiative on S2S (Vitart et al. 2012), the North American Multi-Model Ensemble

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(NMME) subseasonal intercomparison assessment (Robertson et al. 2015), and a U.S. National Research Council agenda on sub-seasonal prediction. As the leading mode of sub-seasonal variability, the MJO is central to these S2S forecast efforts (Waliser et al. 2003b; Zhang 2013). While simple statistical predictions based on time-lagged associations between MJO state and African precipitation are significant at a lead of up to several weeks, advancing beyond simple statistical associations and pushing to longer predictive time horizons requires dynamical models capable of predicting: (1) internal MJO dynamics, including its eastward propagation and generation of Rossby and Kelvin waves, (2) the interaction of MJO with the slowly varying climate modes (ENSO, IOD, etc.) that both modulate the MJO influence and are, in turn, influenced by MJO activity, and (3) the interaction of MJO with synoptic and mesoscale dynamics that lead to S2S variability in affected regions. As Zheng (2013) emphasizes, MJO influence must be understood as a modulation of the occurrence, strength, and spatial patterns of regional climate events, and the nature of this modulation varies with background climate state. It is both a distinct dynamical phenomenon and a bridge between local weather and large scale climate.

The first condition for successful dynamically-based MJO-based predictions—skillful representation of MJO dynamics—has seen significant improvement in recent years. Models are not yet capable of achieving theoretical prediction horizons of 6 or more weeks (Waliser et al. 2003b), but improvements in model resolution, physics, and initialization routines have led to significant skill on 3-4 week horizons (Vitart 2014; Kim et al. 2016; Vitart and Molteni 2010; Vitart et al. 2012; Rashid et al. 2011; Zhang 2013; Neena et al. 2014), albeit with substantial limitations and inconsistencies that are a focus of current model development efforts. As described in Section 5, the ability to simulate MJO interactions with other climate modes has also received concerted attention, and models are improving in their ability to represent, and potentially to predict, MJO behavior under different background climate states. As knowledge

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and model representation of these interactions improves it becomes possible to integrate these interactions into statistical as well as dynamically-based S2S forecasts.

The third challenge—demonstrating skill in the prediction of MJO interactions with relevant synoptic and mesoscale dynamics—is regionally specific, and it is in this area that dedicated studies for Africa are required. In an assessment of MJO teleconnections in hindcasts generated with the ECMWF Integrated Forecast System (IFS Cycle 32r3), Vitart and Molteni (2010) find that the system predicts the general character of MJO teleconnections on Equatorial African precipitation. For boreal winter (November – April), both IFS 16-45 day hindcasts and the ERA-Interim analysis (used as validation data in this study) show positive rainfall anomalies in Equatorial West Africa when the MJO center of convection is in the Western Hemisphere and Africa (RMM phases 8 and 1) and negative anomalies when it is in the Maritime Continent (phases 4 and 5). For Equatorial East Africa, both model and reanalysis show positive anomalies when the center of convection is in the Indian Ocean (phases 2 and 3), and negative rainfall anomalies when it is in the Western Pacific Ocean (phases 6 and 7). Similarly, in boreal summer (June – August) IFS captures teleconnections to the WAM zone that include positive rainfall anomalies when MJO convection is in the Indian Ocean and negative anomalies when MJO is in the Western Pacific. These results are generally consistent with previous studies (e.g., Pohl and Camberlin 2006a; Berhane and Zaitchik 2014; Berhane et al. 2015; Gu 2011), and they demonstrate that a dynamically-based seasonal forecast system is capable both of maintaining a realistic MJO and predicting its impacts on African rainfall. Vitart and Molteni (2010) do note, however, that MJO propagation was generally too slow in this generation of IFS, and that this could explain some discrepancies between hindcasts and observations.

Hogan et al. (2015) examine MJO teleconnection predictions for East Africa in the UK Met Office Unified Model (MetUM). They focus on a short prediction horizon of five days, which means that their results depend almost exclusively on the model's ability to capture dynamics

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that link MJO to East Africa rather than on its representation of MJO propagation. At this timescale, they find that MetUM provides skillful precipitation forecasts and captures the synoptic dynamics of MJO influence that had been identified in earlier studies. MetUM also captures the difference between highland and coastal influences of MJO. Hogan et al. (2015) do note that the model drifts from observation at longer prediction horizons and that the resolution of the model (40km) is not adequate to capture mesoscale dynamics affecting rainfall in the region.

Building on these single model studies, Louvet et al. (2015) study prediction of MJO-related variability in the West African Monsoon in all models participating in The International Grand Global Ensemble (TIGGE) on relatively short (1 to 15 day) prediction horizons. They evaluate forecasted MJO variability by examining correlations between predicted precipitation and Tropical Rainfall Measurement Mission (TRMM) satellite-derived rainfall estimates filtered between 25-90 days. They find strong skill at 10 day forecast horizon in the TIGGE ensemble mean prediction and in the majority of TIGGE models. They do not examine the mechanisms that underlie model predictive performance, but it is interesting to note that the two highest resolution TIGGE models—the European Centre for Medium-range Weather Forecasts (ECMWF) and the Chinese Meteorological Administration (CMA)—show the highest correlation with observation for MJO-associated variability in this evaluation framework. This could indicate that resolution is important for predicting the MJO teleconnection, though that topic was not specifically explored in the study.

This small group of studies signals that high performing dynamically-based S2S forecast systems are able to capture key features of the MJO at time horizons approaching one month, and that they capture teleconnections to selected regions of Africa in a manner that is consistent with reanalysis and available observational datasets. Significant challenges remain, including consistent representation of MJO initiation and propagation at longer prediction time horizons,

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and many processes require further study—for example, representation of Kelvin and Rossby wave dynamics relevant to WAM teleconnections. But the status of S2S forecasts based on MJO has made tremendous progress in recent years.

From an operational perspective, there is potential to leverage improvements in MJO prediction in dynamically-based forecast systems with the time lags inherent in MJO teleconnections to Africa. Statistical associations present at several weeks lead time (e.g., Omeny et al. 2008; Berhane and Zaitchik 2014; Matthews 2004) could, in principle, form the basis of regression models that use dynamically forecasted MJO states as predictors. In this way the 3-4 week MJO prediction in a dynamical forecast system could be chained with a statistically-based forecast with a 2-4 week lag, leading to forecasts of sub-seasonal African rainfall variability at time horizons greater than one month. This approach bears some resemblance to "hybrid" seasonal forecast approaches (Shukla et al. 2014) that make use of both dynamically-based seasonal forecasts and known statistical associations. But the hybrid technique has not yet been applied in the MJO context or for predictions with temporal resolution finer than one month.

Looking beyond S2S timescales, long-term variability or trends in MJO activity could have significant impacts on the character of African precipitation, including changes in extremes, prevailing patterns of intra-seasonal variability, and seasonal rainfall totals. Studies of long-term variability in the MJO have been hampered by two constraints. First, the intra-seasonal nature of the MJO makes it difficult to resolve MJO signals in paleoclimate records, so analyses are generally limited to the period of modern reanalysis datasets (Jones 2016). Second, until very recently GCMs struggled to produce a realistic MJO, and model simulation of the phenomenon is still incomplete and inconsistent (Hung et al. 2013). This makes it difficult to generate model reconstructions of long-term variability or model-based projections of future climate.

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Despite these limitations, a number of recent studies have explored the potential for changes in MJO activity under climate change (Jones 2016; Jones and Carvalho 2011; Takahashi et al. 2011; Liu 2013; Liu et al. 2013; Maloney and Xie 2013; Schubert et al. 2013; Subramanian et al. 2014). These studies adopt different modeling methods, including multi-model analysis of CMIP experiments (e.g., Takahashi et al. 2011) and idealized simulations with a selected GCM (e.g., Maloney and Xie 2013), and none have specifically addressed changes in MJO impacts on Africa. In general, the analyses show large variability in projections between models. Several studies find that the frequency of MJO events increases under greenhouse gas warming (Liu 2013; Jones 2016; Jones and Carvalho 2011), and others find evidence of increased MJO intensity (Schubert et al. 2013; Subramanian et al. 2014). Increases in MJO frequency and/or intensity would be expected to impact intra-seasonal rainfall variability in Africa through the relationships reviewed in Section 3 and 4. Other studies, however, find that Indian Ocean convection increases in a warmer climate but without any clear preference for MJO frequencies (Liu et al. 2013; Maloney and Xie 2013), that there is no consensus in GCM projections for the MJO convective signal (Takahashi et al. 2011), and that changes are sensitive to the spatial pattern of warming (Maloney and Xie 2013). Given these mixed results and the limited realism of MJOs in state of the art GCMs, it is difficult to make any confident conclusion about the influence that warming-induced changes in MJO might have on Africa.

## 7. Future Directions

MJO impacts on Africa were first identified through the creative application of diagnostic tools. The use of appropriate temporal filters, transforms, and spectral analysis were required to isolate and quantify MJO contributions to intra-seasonal variability, first in West Africa and then in other regions. There is still room to improve these diagnostic approaches. For example, diagnostic studies all rely on some kind of MJO indexing system, yet the choice of indexing system is not always consistent across studies. The index systems themselves are also inherently

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limited because they describe a continuous function in terms of a relatively small number of discrete indices. A move towards multi-index analyses across MJO studies will help to make results more comparable, as will continued work to refine the indexing systems themselves. At the same time, the presence of significant event-to-event variability in the MJO suggests that descriptive studies and customized analysis will be required to understand dynamics that are smoothed when events are averaged according to fixed indexing systems (Bellenger and Duvel 2012; Pohl and Camberlin 2014).

MJO studies are also limited by a relatively short historical record. The most widely used indices all depend on satellite-based OLR observations or make use of satellite-era reanalyses, such that the record begins in ~1980. The 35+ year record is sufficient for many studies, but it is limiting when studying MJO interactions with slowly varying climate modes or understanding long-term change (e.g., Pohl and Matthews 2007). The extended MJO reconstruction of Oliver and Thompson (2011) provides a possibility of performing MJO analyses over a significantly longer record (from 1905 forward), which could potentially be quite useful when looking at MJO behavior under ENSO variability. Caution should be exercised when extrapolating the MJO back in time using extended reanalysis products, however, as reconstructed trends can be sensitive to changes in the spatial pattern of assimilated observations (Oliver 2015). For Africa, however, the record in the response variable is also limiting, as historical gauged precipitation data are sparse, and daily gridded rainfall products optimized for Africa start no earlier than 1981—the first year in the CHIRPS record. This is a difficult limitation. Continued work to assemble available historical daily rainfall data for Africa can contribute, but it is also possible that studies of longer term variability will have to rely on reanalysis, models, or dynamical reasoning supported by observations from other regions.

Ultimately, however, observation-based analyses of MJO impacts on Africa need to be supported by robust dynamically-based explanation. Modeling experiments are a powerful tool

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for demonstrating and applying understanding relevant dynamics, but until recent years the failure of most GCMs to produce a realistic MJO limited the approach. Ongoing improvements in GCM representations of the MJO and its teleconnections provide an opportunity to test MJO theory and hypothesized connections to Africa. Interactions between MJO and synoptic dynamics are a particularly important target for such experiments, including the role of AEW in MJO teleconnections to the West African Monsoon, the dynamics of MJO interaction with TTT in Southern Africa, the influence of MJO-induced Kelvin waves (if any) on West African rainfall, and the multi-scale interaction between MJO, large scale modes like ENSO, and synoptic dynamics relevant to East African long and short rains.

There is a synergy between these research needs and goals for MJO-based S2S predictions. Both began with empirical statistical associations and have benefitted from advanced diagnostic techniques, and both now require dynamical models that capture MJO dynamics and teleconnections for the correct physical reasons. The most recent generation of models does show substantial progress in the representation of the MJO, though more work is clearly required. As these models improve it will be important to interrogate simulation results from the perspective of African rainfall variability, and to perform targeted model experiments designed to elucidate the dynamics of MJO-Africa links. The MJO is often described as a bridge between weather and climate. It is also a bridge between powerful atmosphere and ocean dynamics in the Indo-Pacific tropics and some portion of unexplained climate variability experienced in major African rainfall regions. The full extent of these connections, their application to predictions, and the potential for them to change as the planet warms are important subjects for basic and applied research on African climate.

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(Matthews 2004; Maloney and Shaman 2008; Janicot et al. 2009; Pohl et al. 2009; Janicot et al. 2010; Lavender and Matthews 2009; Mohino et al. 2012; Niang et al. 2016; Hung et al. 2013; Leroux et al. 2010; Laing et al. 2011; Ventrice et al. 2011; Berhane et al. 2015; Alaka Jr and Maloney 2012; Gu 2009; Mutai and Ward 2000; Omeny et al. 2006; Berhane and Zaitchik 2014; Oettli et al. 2014; Pohl and Matthews 2007; Tam and Lau 2005; Gushchina and Dewitte 2012; Roundy et al. 2010; Moon et al. 2011; Shimizu and Ambrizzi 2016; Saji et al. 1999; Baldwin and Dunkerton 2001; Hogan et al. 2015; Louvet et al. 2015; Funk et al. 2014; Oliver and Thompson 2011; Berhane 2016)

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## Figure Captions

Figure 1: Annual average rainfall in Africa, 1981-2014, based on Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS; Funk et al., 2014). Boxes approximately indicate the regions reviewed: (1a) West African Monsoon region; (1b) Equatorial West Africa; (3) Equatorial East Africa; (4) Southern Africa. Boxes are approximate because studies differ in their exact definitions. The oval (5) shows the triggering region for African Easterly Waves.

Figure 2: MJO interactions across spatial and temporal scales. The pan-tropical MJO phenomenon influences short-term local rainfall variability through direct convective activity or, more commonly, through modification of synoptic and mesoscale dynamics. It also interacts with hemispheric to global scale climate modes, with phenomena like ENSO influencing and being influenced by MJO activity. Through these interactions MJO can influence rainfall totals on seasonal and longer time-scales. Global climate change acts primarily as a long-term external driver of change in dynamics across spatial scales.

Figure 3: Schematic of the MJO. The convective core (blue) propagates eastward at ~5 m $s^{-1}$. Equatorial Kelvin Waves (K) and Equatorial Rossby Waves (R) are both generated by MJO condensational heating anomalies. Kelvin waves propagate east at phase speed on the order of 30 m $s^{-1}$. Rossby Waves propagate west at speeds on order of 10 m $s^{-1}$. Surface winds converge at the convective core, while elevated winds diverge. The air in front of the MJO convective core tends to be anomalously moist (M), while air behind the convective core is anomalously dry (D).

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Figure 4: RMM phase space diagram, showing an idealized example of two MJO cycles. The first (blue) arises in the Western Indian Ocean and propagates to the Western Pacific before becoming indistinct. The second (gold) arises in the Eastern Indian Ocean and propagates through to the Western Hemisphere. Such a pattern would typically evolve over 2-4 months.

Figure 5: Regression of zero-lag OLR against a customized MJO index that represents active phase MJO over West Africa, for the period 1980-2013. The OLR signal is isolated for (a,b) MJO, (c,d) Rossby wave, and (e,f) Kelvin wave through wavenumber-frequency filtering. Left column (a,c,e) is for spring (March-June) and right column (d,b,f) is summer (July-September). (Figure adapted from Berhane et al. (2015); methodological details provided in that paper.)

Figure 6: Correlation between daily precipitation (CHIRPS) and MJO convective activity in the (a-c) Indian Ocean, for MJO convection centered on $80^0$E; (d-f) Maritime Continent, for MJO convection centered on $120^0$E, during the October-December short rains. NOAA OLR MJO index was used in this analysis. Details provided in Berhane and Zaitchik (2014).

Figure 7: Composites of deseasonalized OLR anomalies (W $m^{-2}$) based on the MJO indexing scheme of Pohl et al. (2007). The left panel shows anomalies for the phase of active MJO convection over tropical Africa, as evident in the OLR signals at the north of the map. The right panel is for MJO suppressed phase over tropical Africa. Shading indicates significance at 95%. Figure is from Pohl et al. (2007) and is reprinted here with permission of the American Meteorological Society.

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Figure 8: Composites of East Africa short rain CHIRPS rainfall (mm/day). (A) RMM Phase 2 MJO and El Niño, (B) RMM Phase 2 MJO and La Niña, (c) RMM phase 2 MJO and non-ENSO, (D) El Niño and non-MJO and (E) La Niña and non-MJO. Figure from Berhane (2016) and reproduced with permission from the author.

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# Highlights

- There is an established body of evidence that links the Madden-Julian Oscillation (MJO) to precipitation variability in Tropical Africa
- The literature addresses numerous regions using diverse methods. This review summarizes and connects these studies in order to assess the state of knowledge.
- Knowledge gaps and research opportunities are identified.

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